

HW3: Data Cleaning Report

- Techniques used (8/8):
- Очистка назв колонок
 - Робота з пропущеними значеннями
 - Зчитування та парсинг дат
 - Зміна типів даних
 - Робота з текстовими даними
 - Обробка числових даних (winsorization)
 - Нормалізація та масштабування
 - Виявлення та видалення дублікатів

Raw shape: (569, 33)
Clean shape: (569, 100)

=== Missing values (raw) ===
No missing values.

=== Missing values (clean) ===
measurement_date 426

=== RAW df.info() ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 33 columns):
Column Non-Null Count Dtype

0 id 569 non-null int64
1 diagnosis 569 non-null object
2 radius_mean 569 non-null float64
3 texture_mean 569 non-null float64
4 perimeter_mean 569 non-null float64
5 area_mean 569 non-null float64
6 smoothness_mean 569 non-null float64
7 compactness_mean 569 non-null float64
8 concavity_mean 569 non-null float64
9 concave_points_mean 569 non-null float64
10 symmetry_mean 569 non-null float64
11 fractal_dimension_mean 569 non-null float64
12 radius_se 569 non-null float64
13 texture_se 569 non-null float64
14 perimeter_se 569 non-null float64
15 area_se 569 non-null float64
16 smoothness_se 569 non-null float64
17 compactness_se 569 non-null float64
18 concavity_se 569 non-null float64
19 concave_points_se 569 non-null float64
20 symmetry_se 569 non-null float64
21 fractal_dimension_se 569 non-null float64
22 radius_worst 569 non-null float64
23 texture_worst 569 non-null float64
24 perimeter_worst 569 non-null float64
25 area_worst 569 non-null float64
26 smoothness_worst 569 non-null float64
27 compactness_worst 569 non-null float64
28 concavity_worst 569 non-null float64
29 concave_points_worst 569 non-null float64
30 symmetry_worst 569 non-null float64
31 fractal_dimension_worst 569 non-null float64
32 measurement_date_raw 569 non-null object
dtypes: float64(30), int64(1), object(2)
memory usage: 146.8+ KB

=== CLEAN df.info() ===
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 569 entries, 0 to 568
Data columns (total 100 columns):
Column Non-Null Count Dtype

0 id 569 non-null int64
1 diagnosis 569 non-null object
2 radius_mean 569 non-null float64
3 texture_mean 569 non-null float64
4 perimeter_mean 569 non-null float64
5 area_mean 569 non-null float64
6 smoothness_mean 569 non-null float64
7 compactness_mean 569 non-null float64
8 concavity_mean 569 non-null float64
9 concave_points_mean 569 non-null float64
10 symmetry_mean 569 non-null float64
11 fractal_dimension_mean 569 non-null float64
12 radius_se 569 non-null float64
13 texture_se 569 non-null float64
14 perimeter_se 569 non-null float64
15 area_se 569 non-null float64
16 smoothness_se 569 non-null float64
17 compactness_se 569 non-null float64
18 concavity_se 569 non-null float64
19 concave_points_se 569 non-null float64
20 symmetry_se 569 non-null float64
21 fractal_dimension_se 569 non-null float64
22 radius_worst 569 non-null float64
23 texture_worst 569 non-null float64
24 perimeter_worst 569 non-null float64
25 area_worst 569 non-null float64
26 smoothness_worst 569 non-null float64
27 compactness_worst 569 non-null float64
28 concavity_worst 569 non-null float64
29 concave_points_worst 569 non-null float64
30 symmetry_worst 569 non-null float64
31 fractal_dimension_worst 569 non-null float64
32 measurement_date_raw 569 non-null object
33 measurement_date 143 non-null datetime64[ns]
34 measurement_date_num 569 non-null int64
35 diagnosis_malignant 569 non-null Int64
36 radius_mean_zscore 569 non-null float64
37 radius_mean_minmax 569 non-null float64
38 texture_mean_zscore 569 non-null float64
39 texture_mean_minmax 569 non-null float64
40 perimeter_mean_zscore 569 non-null float64
41 perimeter_mean_minmax 569 non-null float64
42 area_mean_zscore 569 non-null float64
43 area_mean_minmax 569 non-null float64
44 smoothness_mean_zscore 569 non-null float64
45 smoothness_mean_minmax 569 non-null float64
46 compactness_mean_zscore 569 non-null float64
47 compactness_mean_minmax 569 non-null float64
48 concavity_mean_zscore 569 non-null float64
49 concavity_mean_minmax 569 non-null float64
50 concave_points_mean_zscore 569 non-null float64
51 concave_points_mean_minmax 569 non-null float64
52 symmetry_mean_zscore 569 non-null float64
53 symmetry_mean_minmax 569 non-null float64
54 fractal_dimension_mean_zscore 569 non-null float64
55 fractal_dimension_mean_minmax 569 non-null float64
56 radius_se_zscore 569 non-null float64
57 radius_se_minmax 569 non-null float64
58 texture_se_zscore 569 non-null float64
59 texture_se_minmax 569 non-null float64
60 perimeter_se_zscore 569 non-null float64
61 perimeter_se_minmax 569 non-null float64
62 area_se_zscore 569 non-null float64
63 area_se_minmax 569 non-null float64
64 smoothness_se_zscore 569 non-null float64
65 smoothness_se_minmax 569 non-null float64
66 compactness_se_zscore 569 non-null float64
67 compactness_se_minmax 569 non-null float64
68 concavity_se_zscore 569 non-null float64
69 concavity_se_minmax 569 non-null float64
70 concave_points_se_zscore 569 non-null float64
71 concave_points_se_minmax 569 non-null float64
72 symmetry_se_zscore 569 non-null float64
73 symmetry_se_minmax 569 non-null float64
74 fractal_dimension_se_zscore 569 non-null float64
75 fractal_dimension_se_minmax 569 non-null float64
76 radius_worst_zscore 569 non-null float64
77 radius_worst_minmax 569 non-null float64
78 texture_worst_zscore 569 non-null float64
79 texture_worst_minmax 569 non-null float64
80 perimeter_worst_zscore 569 non-null float64
81 perimeter_worst_minmax 569 non-null float64
82 area_worst_zscore 569 non-null float64
83 area_worst_minmax 569 non-null float64
84 smoothness_worst_zscore 569 non-null float64
85 smoothness_worst_minmax 569 non-null float64
86 compactness_worst_zscore 569 non-null float64
87 compactness_worst_minmax 569 non-null float64
88 concavity_worst_zscore 569 non-null float64
89 concavity_worst_minmax 569 non-null float64
90 concave_points_worst_zscore 569 non-null float64
91 concave_points_worst_minmax 569 non-null float64
92 symmetry_worst_zscore 569 non-null float64
93 symmetry_worst_minmax 569 non-null float64
94 fractal_dimension_worst_zscore 569 non-null float64
95 fractal_dimension_worst_minmax 569 non-null float64
96 measurement_date_num_zscore 569 non-null float64
97 measurement_date_num_minmax 569 non-null float64
98 diagnosis_malignant_zscore 569 non-null float64
99 diagnosis_malignant_minmax 569 non-null float64
dtypes: Int64(1), datetime64[ns](1), float64(94), int64(2), object(2)
memory usage: 445.2+ KB